



BARODA UNIVERSITY

Vadodara | Gujarat | India

"Think Big... Think Beyond"

ITM SLS BARODA UNIVERSITY, VADODARA
B.TECH. SEMESTER-3 (BRANCH – ALL; SCHOOL – SOCSET)
DEPARTMENT OF APPLIED MATHEMATICS (S.O.S.)
COMPUTER ORIENTED NUMERICAL METHODS WITH PYTHON (S2320C1)

TUTORIAL – ERRORS & COMPUTATIONS

Part-A: Multiple Choice Questions

- Q-1. The number of significant digits in the number 1234500 is
(a) 4 (b) 5 (c) 6 (d) 7
- Q-2. The number of significant digits in the number 502.005010 is
(a) 3 (b) 6 (c) 8 (d) 9
- Q-3. Which of the following is not a cause of truncation error
(a) Limiting infinite number of terms in a series to finite number of terms
(b) Chopping off a number
(c) Taking a finite size instead of infinitesimal step size
(d) Limiting infinite number of iterations to a limited number
- Q-4. In a symmetric round-off, the last retained significant digit is rounded by 1, if
(a) the first digit being discarded is greater than or equal to 5
(b) the last digit being discarded is greater than or equal to 5
(c) the last digit being discarded is less than 5
(d) the first digit being discarded is less than 5
- Q-5. If a number is correct to 'n' significant digits, then the relative error is
(a) $\frac{1}{2} 10^n$ (b) $\frac{1}{2} 10^{n-1}$ (c) $\leq \frac{1}{2} 10^{-n}$ (d) $< \frac{1}{2} 10^{n-1}$

Part-B: Theory Questions

- Q-1. What do you mean by the term "error" in numerical methods? List and explain the types of errors occurring in calculations.
- Q-2. Write a short note on Accuracy of Numbers.
- Q-3. List the rules for estimating errors.

Part-C: Numerical Questions

- Q-1. Round off the following numbers correct to four significant figures:
(a) 3.26425 (b) 35.46735 (c) 4985561 (d) 0.70035 (e) 0.00032217 (f) 18.265101
- Q-2. Round off the number 75462 to four significant digits and then calculate the absolute error and percentage error.

Ans: Absolute error = 2, Percentage Error = 0.00265

Q-3. If 0.333 is the approximate value of $1/3$, find the absolute and relative errors.

Ans: Absolute error = 0.0003, Relative error = 0.001.

Q-4. Find the percentage error if 625.483 is approximated to three significant figures.

Ans: Percentage Error = 0.077

Q-5. Find the relative error in taking $n=3.141593$ as $22/7$.

Ans: Relative error = 0.000402

Q-6. The height of an observation tower was estimated to be 47 m, whereas its actual height was 45 m. Calculate the percentage relative error in the measurement.

Ans: Percentage Error = 1.65

Q-7. Suppose that you have a task of measuring the lengths of a bridge and a rivet, and come up with 9999 and 9 cm, respectively. If the true values are 10,000 and 10 cm, respectively, compute the percentage relative error in each case.

Ans: Percentage Error = 0.01 & 10